Day 3 Exercises

Fire, climate, and your own project. Section B is the most important thing in the whole course — it is the one where a convincing map turns out to be wrong by a factor of twenty-four.

A Choosing the right fire dataset

A1

Match each question to the dataset that answers it.

QUESTION	DATASET
A village reports smoke right now. Where is it burning?	
The governor asks how many hectares burned last season.	
Which parts of the burn scar were damaged most badly?	

A2

A report counts 6,494 FIRMS detections in Chiang Mai and concludes that "6,494 km² burned". Give two separate reasons this is wrong — one that makes the figure too high, and one that makes it too low.

A3

NBR uses NIR and SWIR2. Explain what happens to each of those two bands when a forest burns, and therefore what happens to NBR.

B The twenty-four-times error

THE MOST IMPORTANT EXERCISE IN THE COURSE

In Lab 5 you computed dNBR over the whole of Chiang Mai and got **1,594,272 ha** of burned area. MODIS said **67,128 ha**. Both maps look entirely plausible.

B1

Explain the error. Why does dNBR over-report so dramatically in *this particular landscape*, when it works fine in, say, a Canadian conifer forest?

B2

Write the one line of code that fixes it, and explain in one sentence why it works.

B3

After the fix, dNBR reported 67,147 ha and MODIS reported 67,128 ha. Should the fact that they now agree to within 0.03% make you **more** confident, or is it circular reasoning? Think carefully before answering.

B4

Within the confirmed burn scar, 3,702 ha were classified as "unburned" by dNBR. Give a reason the two products can legitimately disagree at the edges even when both are working correctly.

C Climate

C1

Chiang Mai received 2.8 mm of rain in January 2024 and 318.6 mm in August. Chiang Mai's hottest month was **April**, not June. Explain why the hottest month comes before midsummer, and connect it to the fire season.

C2

Your ERA5 temperature values come out as 293.4, 295.8, 298.7. What is wrong and what is the fix?

C3

You want monthly rainfall totals from CHIRPS, whose band is in mm **per day**. A student uses `.mean()` over each month and gets values around 6 mm. What did they do wrong and what should they have used?

D Lab work

D1 · Lab 5 — fire in your province

SEASON	FIRMS PIXELS	BURNED AREA (HA)
2023		
2024		
2025		

Do the two independent measures tell the same story about which year was worst? ▢ yes ▢ no. Why does that matter?

D2 · Severity

SEVERITY CLASS	HECTARES (NAIVE DNBR)	HECTARES (MASKED)
Low		
Moderate-low		
Moderate-high		
High		
Total		

D3 · Lab 6 — climate of your province

QUANTITY	YOUR VALUE
Driest month, and its rainfall	
Wettest month, and its rainfall	
Hottest month, and its temperature	
Ratio of wet-season to dry-season rain	

Does the fire season you measured in D1 line up with the dry, hot months? Write one sentence.

E Mini-project

E1

State your question, in one sentence, before you write any code.

E2

Which datasets will you use? Write the exact IDs from the cheat sheet.

E3

Your answer, as one professional sentence — number, unit, product, year.

E4 · FINAL CHECKLIST

Tick each before you present.

- ☐ The number is in the units I claimed
- ☐ The total is less than the area of my province
- ☐ A second dataset would give roughly the same answer
- ☐ I can name the dataset and year without looking it up
- ☐ My map has a legend and a title
- ☐ I know the biggest weakness of my own answer

E5

What is the biggest weakness of your result? Every result has one. Naming it yourself is what separates a professional from a student.